ARTEMIS PANAGOPOULOU

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EDUCATION

University of Pennsylvania, Philadelphia, PA

Doctor of Philosophy, Computer and Information Science *Aug, 2021 - Aug, 2026*
Research Interests: Natural Language Processing, Computer Vision, Vision-Language Models
Advisor: Mark Yatskar GPA: 3.97/4.0
Thesis: *"Synthesizing Missing Data for Multimodal Models"*

Master of Science in Engineering, Computer and Information Science *Jan, 2018 - Aug, 2020*
Thesis: *"Metaphor and Entailment: Looking at Metaphors Through the Lens of Textual Entailment"*
Advisor: Mitch Marcus GPA: 3.77/4.0

Dual Degree in Artificial Intelligence *Aug, 2015 - Aug, 2020*
Bachelor of Applied Science (BAS), Computer and Cognitive Science.
Thesis: *"Best-First-Model-Merge: From Theory to Implementation and Application"*
Advisor: Mitch Marcus
Bachelor of Arts (BA) Honors, Cognitive Science
Thesis: *"Optical Flow Estimation from Event Based Cameras Using Deep Spiking Neural Networks"*
Advisor: Kostas Daniilidis
Bachelor of Arts (BA) Honors, Philosophy
Thesis: *"On the suitability of Generative Difference Making for addressing challenges in Artificial Intelligence and Robotics."*
Advisor: Lisa Miracchi
Minor in Mathematics GPA: 3.59/4.0

RESEARCH EXPERIENCE

Senior Research Scientist *Sept, 2025 - Present*
Salesforce, Palo Alto CA

- Drive foundational research in multimodal AI and translate cutting-edge research innovations into practical, product-oriented solutions.

Student Researcher *May, 2025 - Nov., 2025*
Google, Mountain View CA

- Engineered and optimized Vision-Language Models (VLMs) for the Augmented Reality (AR) team to enhance real-world spatial understanding.
- Designed and implemented scalable synthetic data generation pipelines for route-tracing on commercial maps, directly addressing data scarcity in multimodal training.

Research Scientist Intern *May, 2023 - August, 2024*
Salesforce, Palo-Alto CA

- Architected and led two projects integrating vision, language, audio, and 3D modalities, resulting in ECCV and CVPR publications.
- Developed and drove a novel Visual Programming framework to enhance the robustness of visual reasoning, resulting in CVPR publication.
- Collaborated on large-scale 3D-vision-text contrastive foundation modeling, enabling scalable pre-training for 3D understanding leading to CVPR publication.

AWARDS AND FUNDING

CTL Graduate Fellowship for Teaching Excellence *August, 2024*
AWS Research Funding for Fair and Trustworthy AI *May, 2023*
Alexa Taskbot Competition Finalist *February, 2022*
President's Engagement Prize *May, 2020*
Dean's List *Aug, 2017 - May, 2020*
Penn Engineering Exceptional Service Award *March, 2019*

PUBLICATIONS

- Panagopoulou, Artemis, Aveek Purohit, Achin Kulshrestha, Soroosh Yazdani, Mohit Goyal. "MapTrace: Scalable Data Generation for Route Tracing on Maps." arXiv preprint (2025).
- Panagopoulou, Artemis, Honglu Zhou, Silvio Savarese, Caiming Xiong, Chris Callison-Burch, Mark Yatskar, Juan Carlos Niebles. "Visual Unit Tests for More Robust Visual Programming', Accepted to Conference in Computer Vision and Pattern Recognition (CVPR) (2025)
- Panagopoulou, Artemis, Le Xue, Honglu Zhou, Silvio Savarese, Ran Xu, Caiming Xiong, Chris Callison-Burch, Mark Yatskar, and Juan Carlos Niebles. *"Contra4: Evaluating Contrastive Cross-Modal Reasoning in Audio, Video, Image, and 3D."*. EMNLP 2025.
- Panagopoulou, Artemis, Le Xue, Ning Yu, Junnan Li, Dongxu Li, Shafiq Joty, Ran Xu, Silvio Savarese, Caiming Xiong, Juan Carlos Niebles. "X-InstructBLIP: A Framework for Aligning X-Modal Instruction Aware Representations to LLMs and Emergent Cross-modal Reasoning", Accepted to European Conference on Computer Vision (ECCV) (2024)
- Panagopoulou, Artemis*, Coby Melkin*, and Chris Callison-Burch. "Evaluating Vision-Language Models on Bistable Images." In Proceedings of The 13th edition of the Workshop on Cognitive Modeling and Computational Linguistics (CMCL 2024) [**Best Paper Award**]
- Xue, Le, Ning Yu, Shu Zhang, Artemis Panagopoulou, Junnan Li, Roberto Martín-Martín, Jiajun Wu, Caiming Xiong, Ran Xu, Juan Carlos Niebles, Silvio Savarese." ULIP-2: Towards Scalable Multimodal Pre-training for 3D Understanding" Accepted to Conference in Computer Vision and Pattern Recognition (CVPR) (2024)
- Chakrabarty, Tuhin, Arkadiy Saakyan, Olivia Winn, Artemis Panagopoulou, Yue Yang, Marianna Apidianaki, and Smaranda Muresan. "I Spy a Metaphor: Large Language Models and Diffusion Models Co-Create Visual Metaphors." In The 61st Annual Meeting Of The Association For Computational Linguistics (ACL). 2023.
- Yang, Yue, Artemis Panagopoulou, Shenghao Zhou, Daniel Jin, Chris Callison-Burch, and Mark Yatskar. "Language in a Bottle: Language Model Guided Concept Bottlenecks for Interpretable Image Classification." Accepted to Conference in Computer Vision and Pattern Recognition (CVPR) (2023)
- Yue Yang*,Artemis Panagopoulou*, Marianna Apidianaki, Mark Yatskar and Chris Callison-Burch. "Visualizing the Obvious: A Concreteness-based Ensemble Model for Noun Property Prediction." Findings of EMNLP 2022.
- Panagopoulou, Artemis, Manni Arora, Li Zhang, Dimitri Cugini, Weiqiu You, Yue Yang, Liyang Zhou, Yuxuan Wang, Zhaoyi Hou, Alyssa Hwang, Lara Martin, Sherry Shi, Chris Callison-Burch, and Mark Yatskar. "QuakerBot: A household dialog system powered by large language models", Alexa Prize TaskBot Challenge Proceedings (2022)
- Yang, Yue, Artemis Panagopoulou, Qing Lyu, Li Zhang, Mark Yatskar, Chris Callison-Burch (2021). "Visual Goal-Step Inference using wikiHow." EMNLP 2021 [**Oral**].

- Yang, Yue, Joongwon Kim, Artemis Panagopoulou, Mark Yatskar, and Chris Callison-Burch. "Induce, edit, retrieve: Language grounded multimodal schema for instructional video retrieval." arXiv preprint arXiv:2111.09276 (2021)
- Chaney, Kenneth, Artemis Panagopoulou, Chankyu Lee, Kaushik Roy, and Kostas Daniilidis. "Self-supervised optical flow with spiking neural networks and event based cameras." In 2021 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), pp. 5892-5899. IEEE, 2021.

PATENT APPLICATIONS

- Systems and methods for multi-modal language models US/18400477
- Systems and methods for training and evaluating multimodal neural network based language models US/18973803
- Systems and methods for visual programming US/18972463

SELECTED INVITED TALKS

Vision Language Models Workshop *Feb, 2025*
Women in Data Science Conference (WiDS), Philadelphia PA
Bridging vision and language: Advances and Challenges *Dec, 2024*
Princeton University, Princeton NJ
Advancing Multimodal AI: Integrating Modalities, Tackling Complex Challenges, and Enhancing Interpretability *Sep, 2024*
University of Pennsylvania, Philadelphia PA

ACADEMIC SERVICE

Served as a reviewer in the following conferences and workshops: **TMLR** (2026), **NeurIPS** (2026), **COLM** (2026), **CVPR** (2025, 2026), **ICLR** (2026,2027), **ACL ARR** (February, August 2024; February, May, October 2025; January, May 2026), **NAACL SRW**, (2025), **COLING**, (2025),**ACL SRW**, (2023, 2024),**ACL**, (2023), **ECCV** (2024)

LEADERSHIP AND ACTIVITIES

Association of Alumnae (AofA) *May 2023 - August 2026*
Advancing Women in Engineering (AWE), Board Member *Aug 2023 - Aug 2025*
Alexa Taskbot Competition Team Lead [Finalist] *Aug 2021 - May 2022*
Women in Computer Science (WiCS) *Jan 2019 - May 2019*